

Install and setup

Graph decomposition

Inspecting Haplotypes with the View Subcommand

Calling Variants

Install and Setup

Fetch the code and test data

```
git clone https://github.com/urbanslug/povu.git --recursive
```

Compile

```
cmake -H. -Bbuild && cmake -build build - -j 3
```

Ensure povu is installed

```
povu -h
```

5 Practicals

Install and setup

Graph decomposition

Inspecting Haplotypes with the View Subcommand

Calling Variants

Graph Decomposition

Run povu decompose on a graph.

```
povu decompose
```

- Creates a tree per component
- Create an output dir to store the PVST

```
povu decompose -i <graph_name>.gfa -o frst
```

PVST File Description

Row label description:

Row label	Description
H	Version number
D	Dummy vertex
F	Flubble

Format description:

H	version			
D	vtx id	vtx label	children	
F	vtx id	vtx label	children	Direction to traverse (L or R)

. is for a null or undefined value
A leaf is a node without children.

Inspect the PVST file

Open the .pvst file and inspect

```
cd frst # switch into the forest directory
```

```
less -s <file>.pvst
```

Hairpin

Yeast dataset

```
povu decompose -i cer.gfa -o frst
```

Compare to Bandage plot

Synchronise

Is this an easier way to reason about the pangenome?

5 Practicals

Install and setup

Graph decomposition

Inspecting Haplotypes with the View Subcommand

Calling Variants

Graph Viewer

View the graph as a collection of haplotypes

```
povu view -i <file>.gfa
```

Synchronise

Compare against bandage plots

Is this an easier way to reason about the pangenome?

⑤ Practicals

Install and setup

Graph decomposition

Inspecting Haplotypes with the View Subcommand

Calling Variants

Generating a VCF with povu

Decompose the graph into a PVST

```
povu decompose -i <file>.gfa -o frst
```

Call variants into VCF with povu call

```
povu call -i <file>.gfa -f frst > x.vcf
```

Both can be done using povu gfa2vcf

```
povu gfa2vcf -i <file>.gfa > x.vcf
```

Datasets

- Organisms
 - Yeast
 - LPA
 - Human subset
- Simulated

Inspect variants using less

```
less -S <file>.vcf
```

Inspect variants using povu

Open the povu TUI with:

```
povu view -i <file>.gfa -c <file>.vcf
```


On own Data?

Can we replicate this on our own datasets?